The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Split Zero-Hidden Selector-Shell Core
Next Benchmark Criterion

Abstract

The current selector-frontier routing note had already fixed the admissible burden order below the theorem forcing the combined response substrate law from the zero-hidden selector generator law $\mathfrak{S}^{\mathrm{sel},0}$: first try to force $\mathfrak{S}^{\mathrm{sel},0}$ itself from still more primitive explicit substrate structure, and if that could not yet be done cleanly, prove a genuinely smaller theorem-level collapse or sharper necessity / uniqueness result strictly inside $\mathfrak{S}^{\mathrm{sel},0}$. The new selector-law collapse theorem has now spent that admissible internal-collapse option: the full selector law $\mathfrak{S}^{\mathrm{sel},0}$ is itself compressed to a smaller split zero-hidden selector-shell core $\mathfrak{C}^{\mathrm{sel,split}}$.

This note records the routing consequence of that new theorem. The next honest benchmark-facing manuscript must therefore go one level below the current split-core frontier: first derive or uniquely force the split zero-hidden selector-shell core $\mathfrak{C}^{\mathrm{sel,split}}$ from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status, or, if that cannot be done cleanly, prove a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside that split core. Another selector-law restatement, combined-response restatement, ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, or another split-core relay that preserves the same split core without shrinking or forcing it further does not count next.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and stronger first-principles promotion language remains paused. The positive result of this note is therefore routing discipline at the new split selector-shell-core frontier.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity is no longer at a stage where the full zero-hidden selector generator law $\mathfrak{S}^{\mathrm{sel},0}$ should be treated as the default unexplained stopping point. The combined-response forcing theorem had already shown that the combined response substrate law $\mathfrak{R}^{\mathrm{comb}}$ is a canonical and necessary reduction of $\mathfrak{S}^{\mathrm{sel},0}$. The new selector-law collapse theorem then spent the admissible internal-collapse move available there: the full selector-law presentation itself carries no extra benchmark-facing freedom beyond a smaller split zero-hidden selector-shell core $\mathfrak{C}^{\mathrm{sel,split}}$.

The present note records the routing consequence of that theorem. It does not reopen the split-core mathematics, and it does not claim a completed first-principles substrate derivation of gravity. It records only which kind of next manuscript would now count as an honest benchmark-facing gain below the current split-core frontier.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It records only which kind of next manuscript would now count as an honest benchmark-facing gain below the split-core frontier.

2. Fixed Inputs

Definition 1 (Split-core-frontier fixed inputs)**.** *The criterion recorded here uses the following fixed inputs.*

1. *The benchmark exact-closure package, which fixes the public one-metric GR target of the repository.*
2. *The benchmark gatekeeping layer, including the post-bridge status correction and the upstream-compression safeguard that blocks same-output deeper-origin relays from counting as new front-line progress once the benchmark-relevant object is unchanged.*
3. *The selector-law forcing theorem, which proves that the combined response substrate law $\mathfrak{R}^{\mathrm{comb}}$ is itself a canonical and necessary reduction of the deeper zero-hidden selector generator law $\mathfrak{S}^{\mathrm{sel},0}$.*
4. *The new selector-law collapse theorem, which proves that the full zero-hidden selector generator law $\mathfrak{S}^{\mathrm{sel},0}$ carries no extra benchmark-facing freedom beyond the smaller split zero-hidden selector-shell core $\mathfrak{C}^{\mathrm{sel,split}}$.*

Remark 1. *The present note does not replace those manuscripts. It records the routing consequence of reading them together under the active safeguard against recursive depth drift.*

3. Why the Split-Core Theorem Is the Frontier

Definition 2 (Benchmark-neutral split-core relay)**.** *A* benchmark-neutral split-core relay *is a manuscript that preserves the same benchmark one-metric output, the same ordinary matter route, the same exact-shell affine nucleus consequence, the same trace-core consequence, the same ambient-package consequence, the same combined-response consequence, the same selector-law consequence, and the same claim boundary while merely restating the full selector law $\mathfrak{S}^{\mathrm{sel},0}$, the combined response law $\mathfrak{R}^{\mathrm{comb}}$, the ambient package $\mathfrak{A}^{\mathrm{bal}}$, the trace core $\mathfrak{C}^{\mathrm{tr}}$, the completion core $\mathfrak{C}^{\mathrm{zr,aff}}$, the exact-shell affine nucleus $\mathfrak{N}^{\mathrm{aff}}$, or the same split selector-shell core $\mathfrak{C}^{\mathrm{sel,split}}$ without shrinking or forcing that split core further.*

Proposition 1 (The selector-law collapse theorem is the live frontier)**.** *At the current stage of the primitive-reservoir observer-reduction line, the theorem collapsing the full selector law* $\mathfrak{S}^{\mathrm{sel},0}$ *to the smaller split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$ *is the live benchmark-facing frontier.*

Proof. That theorem changes benchmark-facing status at current scope because it removes the full selector-law presentation $\mathfrak{S}^{\mathrm{sel},0}$ itself as the live internal stopping point. The current bridge datum is no longer merely a deeper selector package whose combined-response consequence was already known; it is now compressed to the smaller split core $\mathfrak{C}^{\mathrm{sel,split}}$. By contrast, a manuscript that preserves the same benchmark package, the same selector-law consequence, and the same split-core stopping point while merely restating the selector law, the combined response law, or any larger or smaller already-forced package does not alter benchmark-facing status unless it adds a comparably sharp gain. Therefore the live frontier is the selector-law collapse theorem rather than the earlier selector-law forcing theorem or a later same split-core relay. □

Corollary 1 (Status of the full selector law as a next-step target)**.** *Under the current routing safeguard, the full zero-hidden selector generator law* $\mathfrak{S}^{\mathrm{sel},0}$ *is no longer itself the default next benchmark-facing target.*

Proof. The selector-law collapse theorem already proves that $\mathfrak{S}^{\mathrm{sel},0}$ carries no extra benchmark-facing freedom beyond $\mathfrak{C}^{\mathrm{sel,split}}$. Once that is on record, another manuscript that spends the move on the full selector law again does not remove any further benchmark-relevant freedom. The open burden has therefore moved below the full selector-law presentation itself. □

4. The Next Benchmark Criterion

Definition 3 (Admissible next benchmark-facing split-core manuscript)**.** *At the current split-core frontier, a manuscript counts as an admissible next benchmark-facing move only if it respects the following burden order:*

1. *its primary attempt is to derive or uniquely force the split zero-hidden selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$ *from still more primitive explicit substrate structure in a way that does more than merely relocate the unexplained layer deeper;*

2. *if that cannot be done cleanly at current scope, its admissible fallback gain is to prove a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside* $\mathfrak{C}^{\mathrm{sel,split}}$*;*

3. *in either case it preserves the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary, and it introduces neither a second operative metric nor an enlarged low-energy matter law and licenses no stronger first-principles promotion language.*

Theorem 1 (Next honest benchmark-facing task at split-core scope)**.** *The next honest benchmark-facing manuscript at the current split-core frontier should therefore go one level below that frontier: first try to derive or uniquely force the split zero-hidden selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$ *from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status. If that cannot be done cleanly, the admissible fallback is to prove a genuinely smaller collapse, sharper necessity result, or sharper uniqueness result strictly inside that split core. A manuscript that only restates the full selector law* $\mathfrak{S}^{\mathrm{sel},0}$*, only restates the combined response law* $\mathfrak{R}^{\mathrm{comb}}$*, only restates*

the ambient package $\mathfrak{A}^{\mathrm{bal}}$, only restates the trace core $\mathfrak{C}^{\mathrm{tr}}$, only restates the full completion-core presentation $\mathfrak{C}^{\mathrm{zr,aff}}$, only restates the exact-shell affine nucleus $\mathfrak{N}^{\mathrm{aff}}$, or only repeats the same split-core consequence while preserving the same split core without shrinking or forcing it further does not satisfy the criterion.

Proof. By Proposition 1, the live frontier already lies at the theorem that compresses the full selector law to the split core. Therefore a second manuscript below it must improve on that status rather than merely accompany it. A deeper-origin manuscript can do so only if it changes benchmark-facing status by more than depth alone, and the first clean way to do that is to force $\mathfrak{C}^{\mathrm{sel,split}}$ itself from still more primitive explicit substrate structure. If that first option is not yet available cleanly, the routing safeguard allows the fallback of a smaller internal collapse, necessity result, or uniqueness result inside $\mathfrak{C}^{\mathrm{sel,split}}$. If a manuscript simply reproduces the same split-core consequence through another relay while preserving the same split core without shrinking or forcing it further, the routing rule classifies it as benchmark neutral. □

5. What the Next Move Should Not Be

Proposition 2 (Screened next moves)**.** *At the current split-core frontier, none of the following is the default next benchmark-facing move:*

1. *another restatement of the full zero-hidden selector generator law $\mathfrak{S}^{\mathrm{sel},0}$;*
2. *another restatement of the combined response substrate law $\mathfrak{R}^{\mathrm{comb}}$;*
3. *another restatement of the completion-balance ambient dynamics package $\mathfrak{A}^{\mathrm{bal}}$;*
4. *another restatement of the completion-state shell-trace core $\mathfrak{C}^{\mathrm{tr}}$;*
5. *another restatement of the full completion-core graph presentation $\mathfrak{C}^{\mathrm{zr,aff}}$;*
6. *another restatement of the exact-shell affine nucleus $\mathfrak{N}^{\mathrm{aff}}$;*
7. *another split-core relay that preserves the same $\mathfrak{C}^{\mathrm{sel,split}}$ without shrinking or forcing it further;*
8. *a manuscript that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger first-principles promotion language.*

Proof. Items (1)–(7) fail the preceding criterion because they do not directly address the current split-core burden. They revisit either already-forced larger bridge objects, already-forced smaller bridge objects, or the same split-core stopping point rather than removing the remaining split-core freedom or proving a still smaller internal datum inside $\mathfrak{C}^{\mathrm{sel,split}}$. Item (8) violates the fixed claim boundary of the benchmark package and therefore cannot count as a disciplined next step on the active line. The benchmark package remains the exact one-metric GR package already fixed by adoption, so any admissible next manuscript must preserve that boundary while keeping stronger first-principles promotion language paused. □

Remark 2. *This screening statement is not a denial that the selector-law forcing theorem, the combined-response theorem, the ambient-package theorem, the trace-core theorem, or the completion-core theorem matter historically. It is a routing statement: they are not the honest way to spend the next benchmark-facing move from the current split-core frontier.*

6. Conclusion

The positive result of this note is routing discipline at the current split-core frontier. The selector-law collapse theorem remains the live benchmark-facing gain because it already removes the full selector-law presentation as the internal stopping point on the active line. The full selector law therefore no longer sets the honest next burden, and neither do the already-forced combined response law, ambient package, trace core, completion-core presentation, or exact-shell affine nucleus.

The scope remains narrow and honest. The benchmark package is unchanged: the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. Nothing here upgrades adoption to derivation or licenses stronger first-principles promotion language yet.

The next open burden is therefore clear. The next benchmark-facing manuscript should go one level below the current split-core frontier: first try to derive or uniquely force the split zero-hidden selector-shell core $\mathfrak{C}^{\mathrm{sel,split}}$ from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status while preserving the same benchmark one-metric package and claim boundary. If that cannot be done cleanly, the admissible fallback is a genuinely smaller collapse, sharper necessity result, or sharper uniqueness result strictly inside that split core. Another selector-law restatement, combined-response restatement, ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, or another split-core relay that preserves the same split core without shrinking or forcing it further is not the clean next manuscript target at current scope.